VINICIUS MATTOSO

DATA SCIENTIST | MACHINE LEARNING ENGINEER
AI MODELING | DATA SCIENCE SPECIALIST | ARTIFICIAL INTELLIGENCE

CONTACTS & INFORMATION

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Portfólio de IA e codificação

PROFESSIONAL SUMMARY

Engineer specialized in Artificial Intelligence, Machine Learning, and Optimization, with a Bachelor's and Master's degree in Mechanical Engineering. I have solid experience in analyzing and solving complex domain problems, focusing on innovation and practical applications of AI technologies. I am looking for new opportunities to apply my knowledge and deliver intelligent solutions to companies.

SKILLS

- Advanced English

Programming Languages :

- Python, MATLAB, SQL

Tools & Frameworks:

- Scikit-learn, TensorFlow, PyTorch, XGBoost, Prophet, Optuna
- Decision Trees, Neural Networks, Ensemble Models

Model Tracking, Logging & Deployment:

- MLFlow, Docker, FastAPI, Flask

Version Control:

- Git, GitHub

Data Analysis & Manipulation:

- Pandas, NumPy, PySpark
- Time Series Analysis
- Statistical Sample Analysis

Data Visualization & Interfaces:

- Matplotlib, Seaborn, Plotly (Dash), Streamlit

Cloud & Services:

- AWS (S3, EC2, SageMaker, PythonSDK), Azure (ML Studio, PythonSDK)

Processes & Methodologies:

- Technical report writing
- MVP development (development → testing → approval)
- Data preparation and solution implementation
- Development and analysis of new features for models
- Programming and modeling AI-based solutions

EXCHANGE PROGRAM

New York, USA | 02/2015:

- B2 Level - Common European Framework of Reference for Languages (CEFR).
- Kaplan International Languages.

EDUCATION

- **MBA in Computer Vision and Artificial Intelligence for Imaging** | ICA - Rio (In Progress)
- **Postgraduate Degree in Data Science and Analytics** | PUC-Rio 2024
- **MBA in Business Intelligence (BI Master) |** ICA PUC-Rio 2023
- **Master's Degree in Mechanical Engineering (Focus: Petroleum & Energy) |** PUC-Rio 2021
- **Bachelor's Degree in Mechanical Engineering |** PUC-Rio 2019

ADDITIONAL COURSES

- Deep Learning Fundamentals | NVIDIA
- Data Engineering | PUC-Rio
- Introduction to Machine Learning | Coursera
- Introduction to Git and GitHub (Google) | Coursera
- Machine Learning Analyst Bootcamp | XPe
- Python Bootcamp | XPe
- AI II - Artificial Intelligence | PUC-Rio
- Numerical Methods in Python Programming | Udemy
- Python Fundamentals for Data Analysis | Data Science Academy
- Play - Creative Programming of Applications in Python | PUC-Rio
- Introduction to Genetic Algorithms: Theory and Applications | Udemy

PROFESSIONAL EXPERIENCE

Data Science Developer & Researcher

ICA - Puc-Rio | 01/2024 – Present

- Supported the design and implementation of a FastAPI application and a Python library for training ML models in on-premises environments.
- Contributed to the development of submission tools for Azure and AWS clouds, optimizing resource allocation and training efficiency.
- Developed visualization tools using Plotly for lithography characterization, enhancing data interpretation for the team.
- Assisted in developing classification models to distinguish between different lithographies, collaborating with geologists and data scientists.

Teaching Assistant

CCE PUC-Rio | 04/2024 – Present

- Conducted lectures for the BI Master – Business Intelligence Master program.
- Taught courses on Neural Networks, Fuzzy Logic, and Information Retrieval & Utilization (LUI).

Computational Modeling & Optimization Specialist II

LMMP PUC-Rio | 07/2019 - 02/2024

- Developed Python software using Object-Oriented Programming (OOP) for modular and reusable code.
- Performed data preprocessing to handle noise, missing values, and physical inconsistencies.
- Conducted exploratory analysis to evaluate trends, signal freezing, and other phenomena.
- Designed a two-stage modeling approach:
  - Step 1: Modeled physical phenomena.
  - Step 2: Used statistical optimizers to analyze uncertainties based on the first model.
- Prepared technical reports detailing methodologies and results.
- Presented findings at National (INTERPORE 2023) and International (COBEM 2023) Conferences.
- Developed a user interface for interacting with an oil flow simulator.
- Provided support to undergraduate and graduate students on optimization and reservoir simulation.
- Assisted the internal IT team in equipment management and application development.

Previous Roles:

- ***Computational Modeling & Optimization Specialist I***
- ***Research Assistant***

PROJECTS & ACHIEVEMENTS

ICA - Computational Intelligence Lab, PUC-Rio:
- Developed an API to scale Machine Learning model training across different infrastructures.
- Collaborated on the design and implementation of a FastAPI application and a Python library for ML model training in on-premises environments.
- Participated in the development of submission tools for Azure and AWS clouds, improving resource allocation and training efficiency.
- Created visualization tools using Plotly for lithography characterization, enhancing data interpretation.
- Contributed to the development of classification models for distinguishing between different lithographies, working alongside geologists and data scientists.

LMMP PUC-Rio:
- Applied Agile methodology throughout the development cycle.
- R&D Project: Developed a methodology to estimate oil production from each layer of a stratified reservoir using time series data from oil fields, in collaboration with Petrobras.
- R&D Project: Upgraded a non-isothermal reservoir simulator and implemented a set-based optimization method, in partnership with Petrobras.

Chevron (Research Fellow):
- R&D Project: Developed a non-isothermal reservoir simulator and applied it to an optimization method, in partnership with Chevron and PUC-Rio.
- Built the non-isothermal simulator in MATLAB.
- Implemented a Newton-Raphson approximation to estimate reservoir properties.
- Prepared technical reports.
- Presented results at COBEM 2019 (25th International Congress of Mechanical Engineering).

PUC- Rio (Research Fellow):
- R&D Project: Focused on stable production of green nanoparticles and low-impact nanoreactor cleaning techniques:
- Cleaned gold and silver samples to prevent contamination
- Used laser pulses to produce nanoparticles
- Applied spectrophotometry to assess production stability
- Analyzed microscopic images
- Prepared technical reports

PUBLICATIONS

- **2022 Estimation of well production flow rate using Recursive Neural Networks** | Brazilian Congress of Thermal Sciences and Engineering
- **2021 Reservoir characterization using ES-MDA method combining pressure and temperature data |** International Congress of Mechanical Engineering
- **2021 Reservoir Properties Estimation based on Pressure and Temperature Data using ES-MDA |** Ibero-Latin-American Congress on Computational Methods in Engineering
- **2021 Reservoir characterization comparing different ensembles methods |** International Congress of Mechanical Engineering
- **20220 History matching of pressure or temperatura data using ES-MDA for reservoir characterization |** Brazilian Congress of Thermal Sciences and Engineering
- **2020 Reservoir characterization based on transient pressure and temperature data using ensamble-based method |** Rio Oil & Gas
- **2019 Permeability estimation of homogeneous reservoir based on transient temperature and pressure data|** International Congress of Mechanical Engineering